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Arrays

2-pointers

- Two Sum (Sorted) - [Link](#)
- 3Sum - [Link](#)
- 4Sum - [Link](#)

Sliding Window

- Maximum Subarray (Kadane's Algorithm) - [Link](#)
- Minimum Size Subarray Sum -
- Longest Subarray with Sum K - [Link](#)

Sorting and Reordering

- Sort Colors (Dutch National Flag) - [Link](#)
- Move Zeroes - [Link](#)
- Rotate Array - [Link](#)

Merge Sort

- Count Inversions - [Link](#)
- Reverse Pairs - [Link](#)

Voting Algorithm

- Majority Element ($> N/2$ times) - [Link](#)
- Majority Element ($> N/3$ times) - [Link](#)

Hashing

Frequency & Set

- Two Sum (Hashing) - [Link](#)
- Valid Anagram - [Link](#)
- Longest Consecutive Sequence - [Link](#)

Subarray with Prefix Sum + HashMap

- Subarray Sum Equals K - [Link](#)
- Count Subarrays with Equal Number of 0's and 1's
- Longest Subarray with Equal 0s and 1s

Binary Search

Bound Problems

- Lower bound - [Link](#)
- Upper bound - [Link](#)
- Number of Occurrences

Simple BS

- Find Peak Element - [Link](#)
- Search in Rotated Sorted Array - [Link](#)

On Answers

- Aggressive Cows - [Link](#)
- Allocate Minimum Number of Pages - [Link](#)
- Koko Eating Bananas - [Link](#)
- Minimum Days to Make M Bouquets - [Link](#)

2D

- Search in a 2D Matrix - [Link](#)
- Search in a 2D Matrix - II - [Link](#)

Recursion

Subsequence

- Combination Sum - [Link](#)
- Combination Sum II - [Link](#)
- Combination Sum III - [Link](#)

Backtracking

- N Queens - [Link](#)
- Sudoku Solver - [Link](#)

LinkedList

Pointer Techniques

- Reverse a Linked List - [Link](#)
- Detect Cycle in a Linked List - [Link](#)
- Merge Two Sorted Lists - [Link](#)
- Find the intersection point of Y linkedlist - [Link](#)
- Check if LL is palindrome or not - [Link](#)
- Length of loop in LL - [Link](#)

Advanced

- Sort LL - [Link](#)
- Clone a LL - [Link](#)
- Flattenning of LL - [Link](#)

Greedy Algorithms

Interval Scheduling

- N Meetings in One Room - [Link](#)
- Minimum Number of Platforms - [Link](#)
- Merge Intervals

Optimization via Sorting

- Fractional Knapsack
- Job Sequencing Problem - [Link](#)

Sliding Window

Fixed Window Size

- Max Sum Subarray of Size K
- Longest Substring of All 1s After K Flips

Variable Window Size

- Minimum Window Substring - [Link](#)
- Longest Substring Without Repeating Characters - [Link](#)
- Substring with K Distinct Character - [Link](#)

Counting in Window

- Number of Substrings Containing All Three Characters - [Link](#)
- Binary Subarrays With Sum - [Link](#)
- Count number of Nice subarrays - [Link](#)

Stack/Queue

Monotonic Stack

- Next Greater Element - [Link](#)
- Trapping Rain Water - [Link](#)
- Asteroid Collision - [Link](#)
- Remove K Digits - [Link](#)

Cache

- LRU Cache - [Link](#)
- LFU Cache - [Link](#)

Binary Tree

Traversal

- Inorder, Preorder, Postorder - [Link](#)
- Level Order Traversal - [Link](#)
- Zigzag Level Order - [Link](#)
- **Morris Traversals** - [Link](#)

Tree Views

- Vertical Order - [Link](#)
- Top View - [Link](#)
- Bottom View - [Link](#)

Tree Properties

- Height of Binary Tree - [Link](#)
- Diameter of Binary Tree - [Link](#)
- Balanced Binary Tree - [Link](#)

Tree Construction

- Construct Binary Tree from Inorder and Preorder - [Link](#)
- Construct Binary Tree from Inorder and Postorder - [Link](#)
- Construct Binary Tree from Level Order and Inorder

Binary Search Tree

Core BST Operations

- Validate BST
- Kth Smallest Element in BST - [Link](#)
- Delete Node in BST - [Link](#)
- Lowest Common Ancestor - [Link](#)
- Two sum in BST - [Link](#)
- LCA in BST - [Link](#)

Heaps

Heapify

- Heapify Algorithm - [Link](#)

- Heap Sort - [Link](#)
- Build heap from a given Array - [Link](#)
- Implement Min Heap - [Link](#)

Kth Element Problems

- Kth Largest Element - [Link](#)
- Top K Frequent Elements

Graphs

Traversals

- BFS - [Link](#)
- DFS - [Link](#)
- Number of Provinces - [Link](#)
- Number of Islands - [Link](#)

Cycle Detection

- Detect Cycle in Directed Graph - [Link](#)
- Detect Cycle in Undirected Graph - [Link](#)

Topological Sort

- Topological Sort (BFS - Kahn's Algorithm) - [Link](#)
- Course Schedule - [Link](#)
- Detect Cycle in Directed Graph via Topo Sort
- Course Schedule II - [Link](#)
- Alien Dictionary - [Link](#)
- Word Ladder - [Link](#)
- Word Ladder II - [Link](#)

Shortest Path Algorithms

- Dijkstra's Algorithm - [Link](#)
- Shortest Path in DAG - [Link](#)
- Bellman-Ford Algorithm - [Link](#)
- Cheapest flight within K stops - [Link](#)
- Floyd warshall algorithm - [Link](#)

MST

- Disjoint Set - [Link](#)

- Kruskal Algorithm - [Link](#)
- Prim's Algorithm - [Link](#)
- Number of operations to make network connected - [Link](#)
- Accounts merge - [Link](#)
- Number of islands II - [Link](#)
- Most stones removed with same row or column - [Link](#)

Dynamic Programming

1D/2D

- Climbing Stairs - [Link](#)
- House robber - [Link](#)
- Ninja's training - [Link](#)

DP on Grids

- Grid unique paths - [Link](#)
- Grid unique paths II - [Link](#)
- Cherry pickup II - [Link](#)

DP on Stocks

- Best time to buy and sell stock I - [Link](#)
- Best time to buy and sell stock II - [Link](#)
- Best time to buy and sell stock III - [Link](#)
- Best time to buy and sell stock IV - [Link](#)
- Best time to buy and sell stock V - [Link](#)

DP on Subsequences

- Subset sum equals to target - [Link](#)
- Partition a set into two subsets with minimum absolute sum difference - [Link](#)
- 0 and 1 Knapsack - [Link](#)
- Target sum - [Link](#)
- Unbounded knapsack - [Link](#)

LIS

- Longest Increasing Subsequence (Also print it) - [Link](#)
- Longest Bitonic Subsequence - [Link](#)

DP on Strings

- Longest common subsequence - [Link](#)
- Minimum insertions to make string palindrome - [Link](#)
- Edit distance - [Link](#)

MCM DP

- Matrix chain multiplication - [Link](#)
- Burst balloons - [Link](#)

Tries

Basic Trie Operations

- Insert/Search/StartsWith - [Link](#)
- Longest Word With All Prefixes - [Link](#)

Trie-Based Problem Patterns

- Longest Common Prefix - [Link](#)
- Word Search II
- Replace Words